Building the Android Platform: Compile the Kernel

Tired of stock ROMs? Build and flash your own version of Android on your smartphone. This new series of articles will see you through from compiling your kernel to flashing it on your phone.

Many of us are curious and eager to learn how to port or flash a new version of Android to our phones and tablets. This article is the first step towards creating your own custom Android system. Here, you will learn to set up the build environment for the Android kernel and build it on Linux.

Let us start by understanding what Android is. Is it an application framework or is it an operating system? It can be called a mobile operating system based on the Linux kernel, for the sake of simplicity, but it is much more than that. It consists of the operating system, middleware, and application software that originated from a group of companies led by Google, known as the Open Handset Alliance.

Android system architecture

Before we begin building an Android platform, let's understand how it works at a higher level. Figure 1 illustrates how Android works at the system level.

We will not get into the finer details of the architecture in this article since the primary goal is to build the kernel. Here is a quick summary of what the architecture comprises.

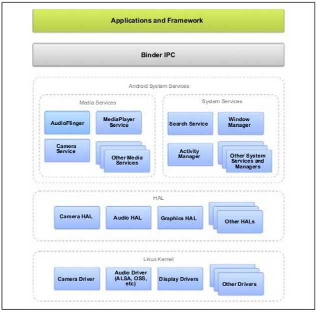

Figure 1: Android system architecture

- *Application framework:* Applications written in Java directly interact with this layer.
- *Binder IPC:* It is an Android-specific IPC mechanism.
- *Android system services:* To access the underlying hardware application framework, APIs often communicate via system services.
- *HAL:* This acts as a glue between the Android system and the underlying device drivers.
- *Linux kernel:* At the bottom of the stack is a Linux kernel, with some architectural changes/additions including binder, ashmem, pmem, logger, wavelocks, different out-of-memory (OOM) handling, etc.

In this article, I describe how to compile the kernel for the Samsung Galaxy Star Duos (GT-S5282) with Android version 4.1.2. The build process was performed on an Intel i5 core processor running 64-bit Ubuntu Linux 14.04 LTS (Trusty Tahr). However, the process should work with any Android kernel and device, with minor modifications. The handset details are shown in the screenshot (Figure 2) taken from the *Setting ->About device* menu of the phone.

Figure 2: Handset details for GT-S5282